MA321 Review 3: Time Series

Name: _____

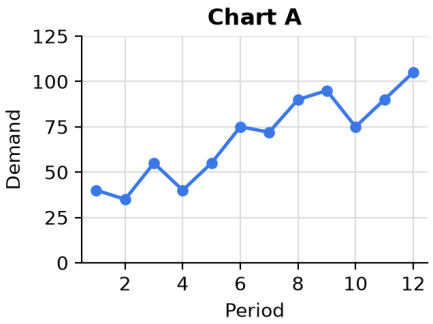
Calculator allowed. No notes, no phone, no internet. Show your work.

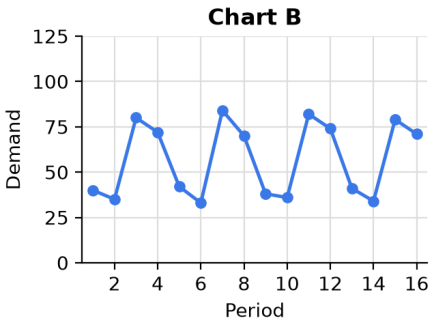
1. Here are the actual demand and the forecast for 3 months. Find the MAPE.

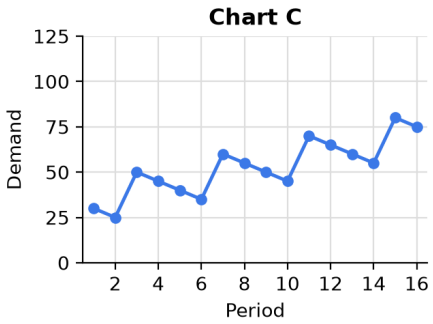
Month	**Actual**	**Forecast**			
1	50	45			
2	80	88			
3	40	50			

MAPE = _____

2. Describe the components in each graph.







Circle True or False.

a) True / False: Chart B is stationary.

b) True / False: Chart A has seasonality.

c) True / False: Chart C has a trend.

d) True / False: Chart A has more noise than Chart C.

e) True / False: Chart C is stationary.

3. Suppose you have an average baseline of 20 units per month. The seasonal indices for the next 6 months are 0.6, 1.2, 0.95, 1.1, 1.4, 0.75.

a) Forecast the next 6 months.

b) Forecast the next 6 months if your average baseline were 20% lower.

4.

a) Forecast 1 has a MAPE of 30% and forecast 2 has a MAPE of 12%. Which one is more accurate?

b) Describe in words how the graph of the forecast compares to the graph of the actuals when the MAPE is 3%.

c) True or False: MAPE can be over 100%.

5. Here are the quarterly averages (composites) from the last 4 years.

	Composite	**Index**
Quarter 1	30	
Quarter 2	45	
Quarter 3	60	
Quarter 4	65	

a) Find the seasonal indices and fill them in.

b) If your average baseline (non-seasonal) prediction for next year is 40 units, what are the seasonalized forecasts for Quarters 1, 2, 3 and 4 next year?